

PRACTICE PROBLEM SET

Evolution, Quantitative Genetics, Molecular Evolution, Population Genetics

(Selection, Mutation, Drift)

Q1. In a population of field mice, fur color ranges from light brown to dark brown. Over several generations, researchers observe that the average fur color becomes darker. However, DNA analysis shows no change in allele frequencies at any known fur-color gene.

- A) Is this population evolving? Explain in one sentence.
- B) Is this population adapting? Explain in one sentence.
- C) Propose one plausible explanation for the observed change in fur color.

Q2. A scientist claims that the human appendix is an adaptation because it produces beneficial immune cells. Another scientist argues it is a vestigial trait.

- A) Define "adaptation" and "vestigial trait."
- B) What evidence would help decide which view is correct?

Q3. A dairy farmer wants to increase milk yield in her herd. The current average milk yield per cow per lactation is 8,500 kg. She selects the top 10% of cows, which have an average yield of 10,000 kg. After one generation of selection, the average yield of the offspring is 9,100 kg.

- A) Calculate the selection differential (S).
- B) Calculate the realized heritability (h^2) from this experiment.
- C) Using this heritability, if the farmer selects cows with an average yield of 10,500 kg in the next generation, what response does she predict?

Q4. A plant breeder is selecting for increased flower size in a population of wildflowers. The initial population mean flower diameter is 4.0 cm. After one generation of selection, the mean flower diameter in the offspring is 4.4 cm. The selected parents had a mean flower diameter of 5.0 cm.

- A) Calculate the realized heritability (h^2) from this experiment.
- B) If the breeder selects parents with mean 6.0 cm in the next

generation, what response does she predict (using the same heritability)?

Q5. A molecular biologist sequences a gene from three closely related frog species. She finds that the number of synonymous differences between Species X and Y is 0.06 per site, and between X and Z is 0.09 per site. The synonymous substitution rate is estimated to be 1.5×10^{-8} per site per year.

Assuming a constant molecular clock, CALCULATE which pair of species (X and Y, or X and Z) shares a more recent common ancestor. Show your calculations.

Q6. For a protein-coding gene, the dN/dS ratio is calculated as 0.15.

A) What does this value indicate about the mode of selection acting on this gene?

B) Would you expect this gene to be highly conserved or rapidly evolving? Explain.

Q7. A pseudogene is identified in the human genome. Its substitution rate is measured to be 1.2×10^{-8} per site per year. The neutral mutation rate (μ) for this region is estimated to be 1.0×10^{-8} per site per year.

A) Are these two rates consistent with neutral theory? Explain.

B) If they differ, propose one possible explanation.

Q8. At a single autosomal locus, the relative fitness values are:

$$AA = 1.0, \quad Aa = 1.0, \quad aa = 0.4$$

The current frequency of the a allele is $q = 0.3$.

A) Is the a allele dominant, recessive, or co-dominant?

B) What type of selection is operating (directional, overdominance, underdominance)?

C) Will the frequency of a increase or decrease in the next generation? Explain.

Q9. In a population of salmon, a recessive allele (d) causes reduced swimming ability. Homozygous recessive individuals (dd) have 40% lower survival than wild-type. The frequency of d is currently 0.1.

A) What is the selection coefficient (s) against dd ?

B) Assuming Hardy–Weinberg proportions before selection, calculate the frequency of d after one generation of selection. Show your work.

Q10. A dominant deleterious allele (H) causes a severe disease. The

mutation rate from the wild-type allele (h) to H is $\mu = 2 \times 10^{-6}$ per generation. The selection coefficient against H (both HH and Hh) is $s = 0.3$.

A) Calculate the equilibrium frequency of H under mutation-selection balance.

B) Why is the equilibrium frequency for a dominant deleterious allele much lower than for a recessive deleterious allele with the same mutation rate and selection coefficient?

Q11. A recessive deleterious allele reduces fitness by 100% (lethal) in homozygotes. The mutation rate from the wild-type allele to the deleterious allele is $\mu = 1 \times 10^{-5}$.

A) Calculate the equilibrium frequency of the deleterious allele.

B) Calculate the frequency of carriers (heterozygotes) at equilibrium.

Q12. At a locus with two alleles (A and a), the forward mutation rate from A to a is $\mu = 3 \times 10^{-5}$, and the backward mutation rate from a to A is $\nu = 1 \times 10^{-5}$. No selection acts at this locus.

A) Calculate the equilibrium frequency of a .

B) If the population starts with A fixed ($p = 1.0$), how many generations would it take to reach equilibrium approximately? (Conceptual answer only.)

Q13. A small, isolated population of birds ($N = 15$) has a neutral allele with frequency 0.05.

A) What is the probability that this allele will eventually become fixed?

B) What will happen to heterozygosity in this population over time? Explain in one sentence.

Q14. A population of fish experiences a severe bottleneck, reducing its size from 10,000 to 20 individuals for one generation, then recovers to 1,000.

A) What is the likely effect on genetic diversity?

B) Would you expect more or less heterozygosity in the recovered population compared to the original? Explain.